

MID TERM EXAMINATION

Subject: computer networks

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 27 | Date: 13-03-2026

Total Questions: 10 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following data structures is most appropriate for managing a print queue where jobs are processed in the order they arrive? [2 Marks]

- a) Stack
- b) Linked List
- c) Queue
- d) Binary Search Tree

2. What is the primary characteristic that distinguishes a singly linked list from a doubly linked list? [2 Marks]

- a) A singly linked list uses less memory per node.
- b) A doubly linked list allows traversal in both forward and backward directions.
- c) A singly linked list can store heterogeneous data types.
- d) A doubly linked list has a fixed size.

3. Which layer of the OSI model is responsible for logical addressing and routing of packets across different networks? [2 Marks]

- a) Data Link Layer
- b) Transport Layer
- c) Network Layer
- d) Application Layer

4. What is the main advantage of using a dynamic array (like ArrayList in Java or std::vector in C++) over a static array? [2 Marks]

- a) Faster element access by index.
- b) Automatically resizes to accommodate more elements.
- c) Stores heterogeneous data types.
- d) Requires less memory overhead.

5. In the context of computer networks, what data structure is commonly used to represent network topologies and facilitate routing algorithms? [3 Marks]

- a) Heap
- b) Graph
- c) Hash Table
- d) Trie

6. Consider a scenario where a server needs to efficiently store and retrieve client IP addresses for connection

management. Which data structure would offer the fastest average-case search and insertion times for this purpose? [3 Marks]

- a) Sorted Array
- b) Balanced Binary Search Tree
- c) Singly Linked List
- d) Hash Table

7. In the TCP/IP model, which protocol operates at the Transport Layer and provides reliable, connection-oriented communication? [3 Marks]

- a) IP (Internet Protocol)
- b) UDP (User Datagram Protocol)
- c) TCP (Transmission Control Protocol)
- d) HTTP (Hypertext Transfer Protocol)

8. Depth-First Search (DFS) and Breadth-First Search (BFS) are graph traversal algorithms. Which of the following applications typically utilizes a BFS-like approach? [3 Marks]

- a) Detecting cycles in a graph.
- b) Finding the shortest path in an unweighted graph.
- c) Topological sorting of a DAG.
- d) Solving mazes.

9. What is the primary method used to prevent overflow in a hash table when a collision occurs? [3 Marks]

- a) Rebuilding the hash table.
- b) Using a different hash function.
- c) Implementing collision resolution techniques like chaining or open addressing.
- d) Limiting the number of items stored in the table.

10. A router's forwarding table needs to quickly match destination IP prefixes to outgoing interfaces. Which data structure is most efficient for this kind of longest prefix matching operation? [4 Marks]

- a) AVL Tree
- b) Splay Tree
- c) B-Tree
- d) Trie (Prefix Tree)

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